

## Assignment 8-4

Name :- Ayush Sharma.

Course - BTECH CSE AI/ML

Roll No :- 2401730128

Code :-

```
import java.io.*;
import java.util.*;
```

```
public class LibrarySystem {
    static class Book implements Comparable<Book> {
        int BookId;
        String title;
        String category;
        boolean isIssued;
```

```
    public Book(int BookId, String title, String author,
                String category) {
        this.BookId = BookId;
        this.title = title;
        this.author = author;
        this.category = category;
        this.isIssued = false;
```

```
    }
    public void markIssued() {
        isIssued = true;
    }
}
```



```
public void marks is Returned() {
    isIssued = false;
}
```

```
public void display Book Details () {
    System.out.println("ID:" + BookId + "Title:" +
        title + "Authors:" + authors + "Category:" +
        category + "Issued:" + isIssued);
}
```

```
}
@Override
public int compareTo (Book b) {
    return this.title.compareToIgnoreCase(b.title);
}
```

```
}
Static Class member {
    int memberId;
    String name;
    String email;
    List<Integer> issuedBooks = new ArrayList<>();
}
```

```
public member (int memberId, String name, String email) {
    this.memberId = memberId;
    this.name = name;
    this.email = email;
}
```

```
public void addIssued Book (int BookId) {
    issuedBooks.add(BookId);
}
```

```
public void return Issued Book (int BookId) {
    issuedBook.remove(Integer.valueOf(BookId));
}
```



```

}
    public void displaymemberDetails() {
        System.out.println("Id:" + member.Id + " "
            + "Name " + name + " " + "Email " + email + " "
            + "Issued Books:" + IssuedBooks);
    }
}

```

```

static class Librarymanager {
    map<Integer, Book> Books = new HashMap<>();
    map<Integer, Member> member = new HashMap<>();

    File BookFile = new File("Books.txt");
    File memberFile = new File("member.txt");
}

```

```

public void loadFromFile() {
    try {
        if (!BookFile.exists()) BookFile = createNewFile();
    }
}

```

```

    if (!memberFile.exists()) memberFile = createNewFile();

    BufferedReader br = new new BufferedReader(new
        FileReader(BookFile));
}

```

```

    String line;
    while ((line = br.readLine()) != null) {
        String[] p = line.split(",");
        Book b = new Book(Integer.parseInt(p[0]),
            p[1], p[2], p[3], p[4]);
        b.Issued = Boolean.parseBoolean(p[4]);
    }
}

```

```

    br.close();
}

```



```

BufferedReader br2 = new BufferedReader(new
    FileReader(memberFile));
while ((line = br2.readLine()) != null) {
    String[] p = line.split(",");
    member m = new member(Integer.parseInt(
        p[0]), p[1], p[2]);
    member.put(m, memberId, m);
}
br2.close();
}
catch (Exception e) {
    System.out.println("Error loading files.");
}
}

public void SaveToFile() {
    try {
        Buffered Writer bw = new Buffered Writer (new File
            writer(bookFile));
        for (Book b : books.values()) {
            bw.write(b.bookId + ", " + b.title + ", " +
                b.author + ", " + b.category + ", " + b.
                isIssued);
            bw.newLine();
        }
        bw.close();
    }
    Buffered Writer bw2 = new Buffered Writer (new
        FileWriter(memberFile));
    for (member m : member.values()) {
        bw2.write(m.memberId + ", " + m.name + ", "
            + m.email);
        bw2.close();
    }
}

```



```

    catch (Exception e) {
        System.out.println("Error saving file");
    }
}

```

```

public void addBook() {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter BookId");
    int id = sc.nextInt();
    sc.nextLine();
}

```

```

    System.out.print("Enter Title:");
    String title = sc.nextLine();
}

```

```

    System.out.print("Enter author:");
    String author = sc.nextLine();
}

```

```

    System.out.print("Enter category:");
    String category = sc.nextLine();
}

```

```

Book b = new Book(id, title, author, category);
book.put(id, b);
saveToFile();
System.out.println("Book added Successfully");
}

```

```

public void addMember() {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter MemberId:");
    int id = sc.nextInt();
    sc.nextLine();
}

```



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```
System.out.print("Enter Name");  
String name = sc.nextLine();
```

```
System.out.print("Enter Email");  
String email = sc.nextLine();
```

```
Member m = new Member(id, name, email);  
members.put(id, m);  
saveToFile();
```

```
System.out.println("Member Added Successfully");
```

```
public void IssueBook()  
{  
    Scanner sc = new Scanner(System.in);
```

```
    System.out.print("Enter Book ID");  
    int bid = sc.nextInt();
```

```
    System.out.print("Enter Member ID");  
    int mid = sc.nextInt();
```

```
    if (!Books.containsKey(bid) || !members.containsKey(mid))
```

```
    {  
        System.out.println("Invalid ID's");  
        return;
```

```
    }
```

```
    Book b = Books.get(bid);  
    Member m = members.get(mid);
```

```
    if (b.isIssued)
```

```
    {  
        System.out.println("Book already issued!");  
        return;
```

```
    }
```



```

        b.mark As Issued ();
        m.add Issued Book (bid);
        Save to file ();
        System.out.println ("Book Issued Successfully");
    }

    public void returnbook () {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter Book Id:");
        int bid = sc.nextInt ();
        System.out.print ("Invalid Book Id");
        return;
    }

    Book b = Books.get (bid);
    b.mark As Returned ();
    SaveTofile ();
    System.out.print ("Book Returned Successfully");

    public void Search Books () {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter Title to Search:");
        String t = sc.nextLine ();

        For ( Book b : Books.values () ) {
            if (b.title.equals (t)) {
                b.display Book Details ();
            }
        }
        return;
    }

    System.out.print ("Book not found");

    public void sort Books () {
        List <Book> list = new ArrayList <> (Books.values ());
    }

```



```

        collection.sort(list);
        System.out.println("Books Sorted by title");
        for (Book b : list) {
            b.displayBookDetails();
        }
    }
}

```

```

public static void main (String [] args) {
    LibraryManager lm = new LibraryManager();
    lm.loadFromFile();
    Scanner sc = new Scanner(System.in);
    int choice;
    do {
        System.out.println("\n City Library Digital  
management System ---");

```

```

        System.out.println("1. Add Book");
        System.out.println("2. Add member");
        System.out.println("3. Issue Book");
        System.out.println("4. Return book");
        System.out.println("5. Search Book");
        System.out.println("6. Sort Books");
        System.out.println("7. Exit");
        System.out.println("Enter choice:");
        choice = sc.nextInt();

```

```

    switch (choice) {
        case 1: lm.addBook(); break;
        case 2: lm.addMember(); break;
        case 3: lm.issueBook(); break;
        case 4: lm.returnBook(); break;
        case 5: lm.searchBook(); break;

```



case 6: `lim. SortBook(); break;`  
case 7: `lim. SaveToFile(); break;`  
default: `System.out.println("Invalid choice");`

`}`  
`} while (choice != 7);`  
`System.out.println("Thank you for using the System.");`

`}`

`}`